# ClipQuest

Iteration overview for the 36-hour challenge

> **Project:** ClipQuest
> **Repository:** https://github.com/404-Waifu-Not-Found/HHCC2026
> **Live product:** https://clipquest.ccwu.cc

## Submission summary

This document describes the work completed during the event and the difference between the pre-event foundation and the post-event iteration.

## Pre-event foundation

Before the event, the project had only a short written README describing the initial product idea. There was no implemented product flow, completed frontend, quiz engine, persistence layer, browser extension, native client flow, or production deployment to evaluate.

The event iteration turned that written concept into the codebase and working product submitted here.

## What changed during the event

The team built the full product in this project. The work was iterative: each stage addressed a concrete product risk discovered in the previous stage.

| Iteration | Problem or goal | Improvement delivered |
| --- | --- | --- |
| Product foundation | A written idea needed a usable learning flow. | Built the web app, authentication, URL import, lesson preview, quiz flow, feedback, completion state, and Library. |
| Evidence-grounded generation | AI questions needed reliable lesson context. | Added caption-only public YouTube acquisition, normalization, structured DeepSeek generation, strict schemas, and explicit caption failure states. |
| Progressive learning | Waiting for a complete | Added question-first progressive generation so the first |

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| | quiz felt slow. | validated question opens the attempt while the rest continue in bounded batches. |
| Correctness and recovery | A malformed question could break an attempt. | Added ordinal validation, duplicate checks, repair of the first missing question, idempotent imports, cancellation, timeouts, and safe recovery. |
| Mastery loop | A result alone did not create repeatable learning. | Added Library history, color-coded mastery, review state, progress bars, due reviews, saved lessons, and practice presentation. |
| Cross-platform access | The experience needed to work beyond one browser screen. | Added a Chrome extension bridge plus native iOS and Android paths sharing the local quiz engine and privacy boundary. |
| Product polish | The first UI needed stronger hierarchy and feedback. | Redesigned cards, hover/theme states, progress treatment, streaming feedback, empty/error states, responsive layouts, accessibility, and reduced motion. |
| Release readiness | A working feature needed repeatable delivery. | Added contracts, migrations, API routes, tests, asset verification, extension packaging, Wrangler deployment, health checks, and release docs. |

## Pre-event versus post-event

| Area | Pre-event | Post-event |
|---|---|---|
| Product state | Short README only | Deployed web product with browser extension and native implementation paths |
| Input | Product idea | Public YouTube URL validation, metadata preview, and caption acquisition |
| AI | No implemented AI flow | Learner-owned DeepSeek key used locally for structured question generation |
| Learning | No quiz experience | Multiple-choice, true/false, and short-answer quizzes with immediate feedback |
| Progress | No persistence or mastery model | Authenticated attempts, ordered storage, mastery ranks, review state, and Library |

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|---|---|---|
| Reliability | No validation or recovery | Contract validation, bounded retries, repair, cancellation, timeouts, and idempotency |
| UX | No interface | Responsive UI with light/dark themes, motion, accessibility, and clear states |
| Platforms | No working client | Web, Chrome extension bridge, iOS path, and Android path sharing core logic |
| Delivery | No deployment | Cloudflare Worker and static assets deployed at clipquest.ccwu.cc |
| Verification | No test suite | Automated contract, API, app, engine, extension, UI, build, and asset checks |

## Improvement logic and scope

1. **Ground the content.** Use public captions rather than invented lesson content. Unsupported or captionless sources fail explicitly.
2. **Protect correctness.** Validate every question before it becomes learner-visible or persistent. Accepted questions are never silently replaced.
3. **Reduce waiting.** Progressive generation makes question 1 available quickly while preserving ordered completion.
4. **Make progress meaningful.** Results become mastery, review, and Library state.
5. **Make the product usable.** Responsive layouts, theme parity, accessible controls, visible states, and motion-safe feedback are part of the feature.
6. **Ship with evidence.** Automated checks and production deployment are included; unfinished features remain clearly gated.

The current release intentionally hides the Workplace AI chat navigation for all users while that feature remains in development. The route and source remain behind a release gate for future work.

## Team decisions and responsibilities

- Core product concept and UX flow
- Feature scope and prioritization
- Project architecture and file structure
- Caption-only privacy boundary

- Key algorithms and validation/recovery logic
- Mastery and review model
- Testing strategy and CI/release verification
- Visual direction, interaction design, and accessibility expectations

# AI usage disclosure

### Usage cost

**$240 in API-token usage across all tools**

### Tools used

- GitHub Copilot
- Claude Code CLI

All implementation code for the event iteration is written in this project repository.

### Team decisions and concepts

- Project architecture and file structure
- Core product concept and UX flow
- Feature scope and design decisions
- Key algorithms and logic design
- Testing strategy and CI pipeline setup

### AI as an auxiliary tool

AI was used only for boilerplate and repetitive code, syntax lookup and autocomplete, refactoring and code formatting, documentation drafting, and debugging assistance.

**All key decisions, concepts, and creative direction were developed entirely by the team. AI served as an auxiliary coding tool only.**

# Recommended judge submission

1. **Project name:** ClipQuest

2. **Iteration code:** https://github.com/404-Waifu-Not-Found/HHCC2026
3. **Iteration overview:** ITERATION-OVERVIEW.pdf
4. **Presentation/demo:** https://clipquest.ccwu.cc
5. **AI disclosure:** Include the AI usage disclosure section above

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The repository contains the complete event implementation. The README and documentation provide additional architecture, privacy, release, and QA context.